

James Ouyang

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Education

University of Waterloo
B.Sc. Computer Science

Sept 2020 – Apr 2025

Experience

Computer Systems Analyst
Quantile Health Inc.

Offer Accepted – Start Date: Sept 29, 2025

- Analyze and evaluate computer systems and workflows to recommend improvements in efficiency, scalability, and reliability.
- Design specifications for software and data systems supporting healthcare analytics and clinical applications.
- Collaborate with stakeholders to document requirements and translate business needs into technical system designs.
- Assess integration of databases, APIs, and cloud infrastructure; recommend enhancements for performance and security.
- Support implementation teams by preparing technical documentation, testing procedures, and optimization strategies.

Quantitative Analyst
Trexquant Investment LP

Aug 2024 – Dec 2024

- Analyzed research workflows and designed an integrated pipeline combining LLM components for summarization, literature review, and visualization.
- Evaluated and optimized 881 fundamental **alpha signals**, improving overall information ratio by **20 %**.
- Designed and tested forecasting systems (LightGBM/XGBoost) for a 2,000-stock universe, validating performance with live **Sharpe > 2**.

System Engineer
ODAIA

Jan 2024 – May 2024

- Assessed requirements and designed a knowledge-retrieval system (LangChain + GPT) to interpret pharmaceutical sales forecasts; achieved **92 % analyst adoption**.
- Implemented and validated a retrieval-augmented pipeline for structured/unstructured data (PDF, CSV, SQL), reducing query latency to sub-second.
- Re-engineered the ML serving architecture (FastAPI + ECS), improving cost efficiency by **65 % (\$100K annually)**.

Data Systems Analyst
Genellipse

Sep 2022 – Dec 2023

- Designed and deployed an information retrieval system (LlamaIndex + FAISS) to synthesize SEC filings, insurance policies, and clinical guidelines.
- Analyzed system accuracy issues and engineered a custom reranker to reduce false positives by **30 %**.
- Built and optimized a data ingestion pipeline (SQL + Airflow) for healthcare claims and EHR data; increased throughput by **246 %**.

Technologies

Languages: Python, C++, C, SQL, Bash

System Tools & Libraries: LangChain, LlamaIndex, PyTorch, TensorFlow, LightGBM, XGBoost, FastAPI

Data & Infrastructure: AWS (S3, RDS, Lambda, ECS), Docker, Terraform, Snowflake, Pinecone, Weaviate